

Multimodal Fake News Classification

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1 Introduction

This project explores the multimodal exploration of fake news using the Fakeddit dataset, which provides three types of labels: 2-way, 3-way, and 6-way. We focus primarily on the relationships between features and the 2-way labels (True vs. False) and the 6-way labels (0: True, 1: Satire, 2: False Connection, 3: Imposter Content, 4: Manipulated Content, 5: Misleading Content).

An initial evaluation of missing data shows that the feature `linked_submission_id` has the highest proportion of missing values (around 70%), while `domain`, `num_comments`, and `upvote_ratio` each have about 30% missing values. Most other columns have little to no missing data.

2 Label Distributions Across Splits

Analyzing the data reveals a clear class imbalance which remains same across the training, validation, and test sets, indicating that these splits were created using stratified sampling. In the 2-way setup, the class distribution is approximately 2:3. In the 3-way setup, the imbalance becomes much more pronounced, with about 60% of the data belonging to a single dominant class. In the 6-way setup, the distribution is more diverse, with several minority classes having very few samples and a few majority classes dominating. The overall imbalance becomes more severe as the number of classes increases. This suggests that learning may become harder with more classes and could require strategies like class weighting or data augmentation.

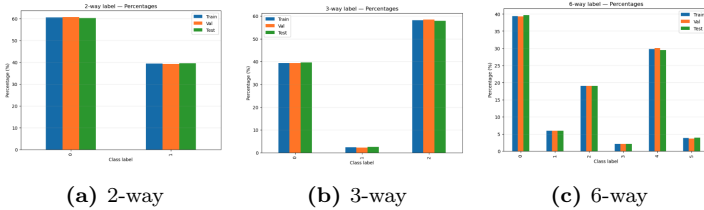


Figure 1: Label distributions across train/validation/test.

We will next evaluate key features to examine their distributions to understand if these features differ enough across categories to be useful for prediction.

3 Feature Distributions by Labels

3.1 Title Length Density

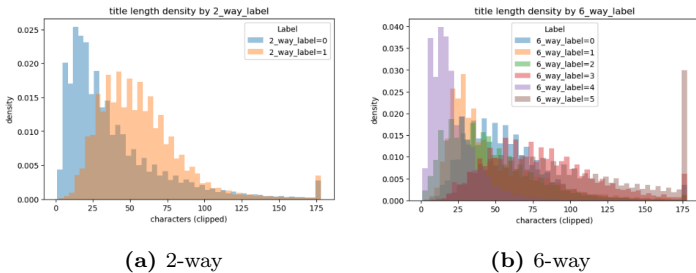


Figure 2: Title length density.

Title lengths are generally short, with most clustered below 50 characters, and their density gradually declining as length increases. In the 6-way labels, some categories (such as `6_way_label1`) tend to have notably shorter titles, while others are more spread out. In contrast, the 2-way label distributions overlap more but still show slightly longer titles for one class. This difference in title length distributions suggests that title length could be a useful predictive feature.

3.2 Author

Among the top authors by class labels, most are strongly associated with only one label, showing very little diversity in the types of posts they make.

In the 2-way label setup, nearly all top authors post exclusively class 0 content, with only one author showing a mix of both classes. In the 6-way label setup, this pattern is even clearer—each author is almost entirely tied to a single label.

This indicates that author identity aligns strongly with specific labels and could serve as a useful signal for prediction.

3.3 Title Word Count

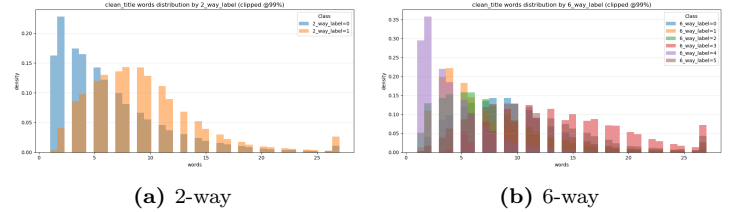


Figure 3: Title word-count distributions.

In the 2-way setup, titles from class 0 are generally (2-5 words) shorter, while class 1 titles are noticeably longer (6-19 words), in line with our finding from title length density. In the 6-way setup, the variation becomes more nuanced: some classes are highly concentrated around very short titles (1-3 words), while others have broader distributions or much longer titles extending beyond 15 words. Although there is overlap among several classes, these distinct length profiles for certain classes indicate that title word count can still provide useful complementary information.

3.4 Domains

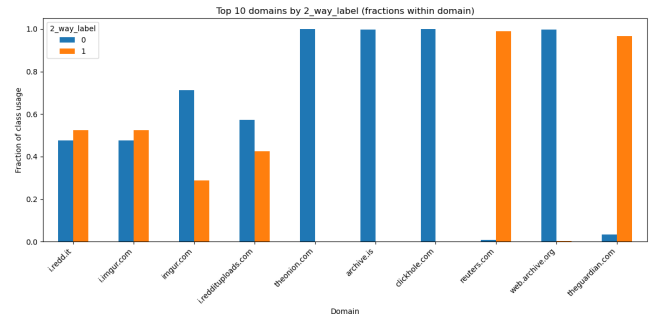


Figure 4: Top domains by label (2-way).

Evaluating domains, we see that there are certain domains that always show a single label, whereas some have a mixture, so we can hypothesize that there are certain domains made to operate only on fake news with either no or a small percentage

of other types of news (maybe in an attempt to mask their actual activity).

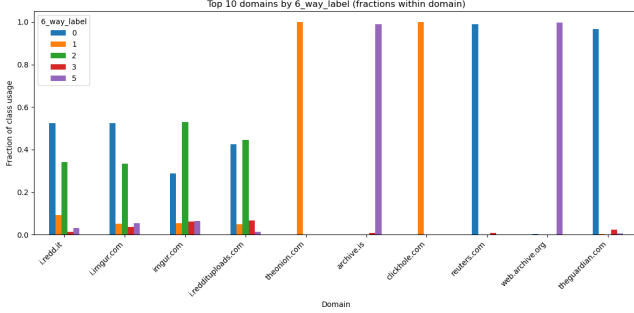


Figure 5: Top domains by label (6-way).

Trusted news sites are almost entirely class 1 in the 2-way setup, while satirical or archived domains are dominated by class 0. In the 6-way setup, this pattern becomes even clearer, with domains showing near-exclusive association with single classes, indicating that domain is a highly discriminative feature. To further analyze this, we investigate the entropy per class.

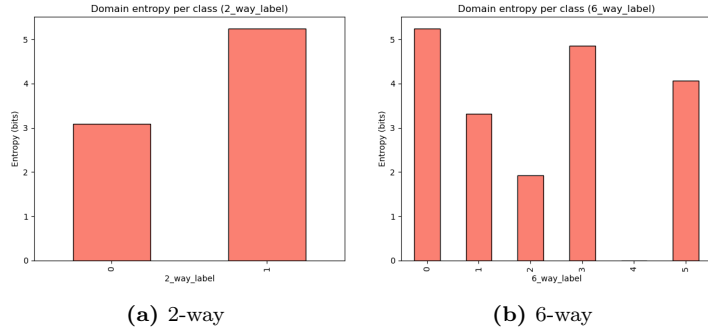


Figure 6: Domain entropy per class.

This reveals that some classes draw from a much narrower set of domains than others. In simple words, low entropy means that most posts of that class come from just a few domains, while high entropy means they are spread across many different domains. A limitation here is missing values, specially for class 4 in the 6-way label which contains all "NAN" values.

3.5 Engagement Correlations

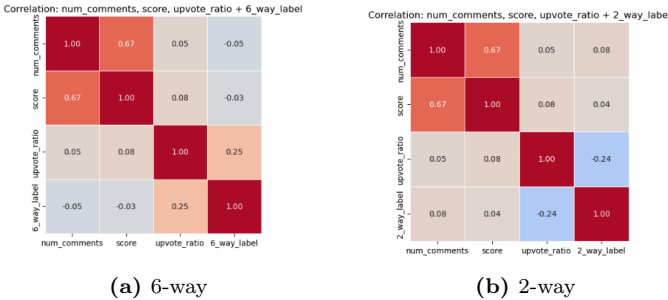


Figure 7: Engagement metrics vs labels.

The correlation heatmaps of key numerical engagement features show that `num_comments` and `score` are moderately correlated with each other (≈ 0.67), which is expected since posts with more comments often receive higher scores. However, their correlation with the labels is very weak, indicating that these metrics do not vary meaningfully across classes. `upvote_ratio` shows a slight positive correlation with the 6-way label (0.25) and a slight negative correlation with the 2-way label (-0.24),

suggesting that certain classes may receive somewhat higher or lower upvote ratios, though this relationship is still weak. Overall, these engagement metrics show minimal direct association with class labels, though they could still contribute small complementary signals when combined with other features.

4 Image Properties

Using a subset of images, we see that visual properties vary across classes in subtle but noticeable ways. In the 2-way setup, class 1 images are generally slightly larger, brighter, and more visually complex than class 0, which appears more compact and less textured. In the 6-way setup, this variation becomes clearer: classes 3 and 5 show higher brightness and complexity, while classes 0 and 2 are generally darker, simpler, and slightly smaller. These trends suggest that basic image features—like size, brightness, and complexity—could provide useful complementary signals for distinguishing certain classes.

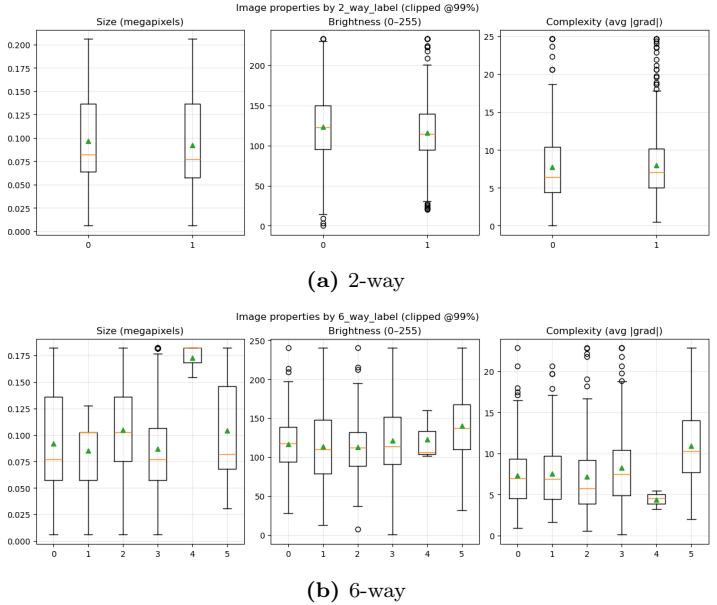


Figure 8: Image size, brightness, and complexity by label.

5 Identified Challenges

A key challenge is the class imbalance, which becomes more severe as we move from 2-way to 6-way labels, making it harder for models to learn minority classes effectively. Additionally, there are substantial missing values in some features, and several features—such as engagement metrics—show only weak correlation with labels, limiting their predictive power. There is also high domain concentration within certain classes, which may cause models to overfit to specific domains rather than learning generalizable patterns.

6 Conclusion

This exploration shows that while multimodal signals (text, metadata, images) offer valuable cues, imbalanced labels, missing data, and domain-specific biases pose major hurdles. Addressing these challenges through data augmentation, class weighting, and domain-invariant modeling will be crucial to build robust and generalizable fake news detection systems.